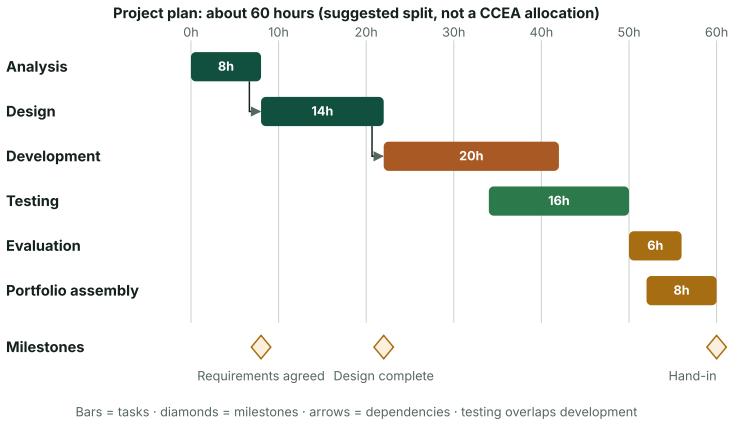
Understanding the task: case study, planning and authentication

AD01 WORKSHEET

Name	Class	Date	Score

I can: Describe what the A2 2 task involves and how the portfolio is organised · Produce a realistic Gantt chart for a 60-hour project · Explain the rules for authentication and how to acknowledge help and sources



Bars = tasks · diamonds = milestones · arrows = dependencies · testing overlaps development

Use this diagram to help you: A Gantt chart for a 60-hour database project showing six tasks, overlapping testing, dependencies and three milestones

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Case study**
- ☐ **Gantt chart**
- ☐ **Milestone**
- ☐ **Dependency**
- ☐ **Critical path**
- ☐ **Authentication**

A Confirmation by the candidate and teacher that coursework is the candidate's own work, produced under the required conditions.

B A planning chart showing each task as a horizontal bar against a timeline, with durations, overlaps and dependencies.

C A relationship between tasks in which one task cannot start until another has been completed.

D The longest sequence of dependent tasks in a project; a delay to any task on it delays the whole project.

E A significant checkpoint in a project, such as the completion of a stage, used to monitor progress.

F A scenario pre-released by CCEA describing an organisation's problem, which candidates analyse and solve with a database application.

B Fill the gaps

Use the word bank · 6 marks

case study | teacher | database | Gantt chart | milestone | authenticated

A2 2 is assessed by your _____ and moderated by CCEA. You build a relational _____ application in response to a pre-released _____. A _____ shows tasks against time, and

a _____ marks an important checkpoint. All help and sources must be acknowledged so the work can be _____.

C True or false?

Tick one box · 6 marks

	True	False
A2 2 evidence is submitted as a single PDF portfolio.	☐	☐
The case study is the same every year, so last year's notes can be reused.	☐	☐
A dependency means one task cannot start until another has finished.	☐	☐
You do not need to acknowledge online tutorials if you changed the wording.	☐	☐
Updating your Gantt chart after a delay is good evidence of project management.	☐	☐
The moderator can view your database file, so screenshots are optional.	☐	☐

D Exam-style questions

20 marks

1 State what is produced in the A2 2 Application Development task and how the evidence is submitted. [2]

2 Explain why a Gantt chart is useful when planning a 60-hour project. [3]

3 Define the term milestone and give an example from a database project. [2]

4 Describe two things a candidate must do to ensure their coursework can be authenticated. [4]

5 Explain why requirement numbers should be used throughout the portfolio. [3]

6 A candidate plans to spend 40 of the 60 hours on development and leave testing and evaluation until the last week. Evaluate this plan. [6]

E Quick quiz

Circle one answer · 5 marks

1 What does the A2 2 task require you to produce? [1]

A A game written in C#
B A relational database application
C A website
D A written essay

2 Which shows tasks as bars against a timeline? [1]

A Data flow diagram
B ER diagram
C Gantt chart
D Wireframe

3 What is a milestone? [1]

A A key checkpoint in the project
B A type of query
C A field in a table
D An error message

4 Which of these is malpractice? [1]

A Citing a tutorial you used
B Signing the authentication declaration
C Keeping a project log
D Submitting another pupil's design as your own

5 Where should you check the marking criteria? [1]

A A revision guide from 2017
B The current CCEA marking grid
C A friend's portfolio
D The software help menu

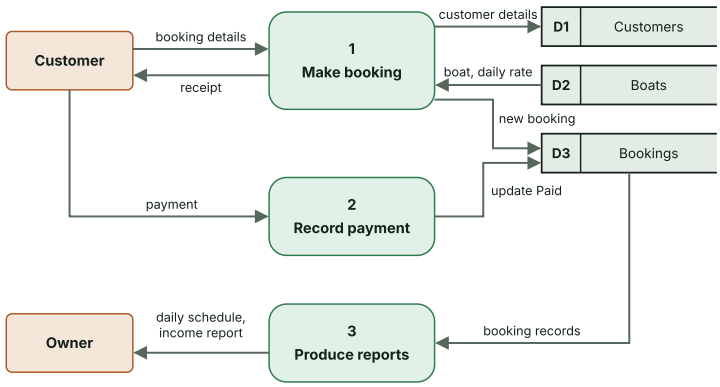
A2 2: APPLICATION DEVELOPMENT · APPLICATION DEVELOPMENT PROJECT

Analysis: client, current system and requirements

AD02 WORKSHEET

Name	Class	Date	Score

I can: Plan and record a client interview that gathers useful detail · Write numbered, measurable functional and non-functional requirements · Draw context and level 1 data flow diagrams of the proposed system



Rectangle = external entity · rounded box = process · open box = data store · arrow = data flow

Use this diagram to help you: A level 1 data flow diagram for the proposed Erne Boat Hire system with two external entities, three processes and three data stores

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Requirements specification**
- ☐ **Functional requirement**
- ☐ **Non-functional requirement**
- ☐ **Measurable requirement**
- ☐ **Data flow diagram**
- ☐ **Context diagram**

A A top-level data flow diagram showing the whole system as one process together with its external entities and data flows.

B A requirement with a specific, testable target so that it can be judged objectively as met or not met.

C A diagram that models how data moves between external entities, processes and data stores in a system.

D A statement of what the system must do, such as the data it stores, the processes it carries out or the outputs it produces.

E A statement of how well the system must perform, covering qualities such as usability, performance, security or reliability.

F A numbered, agreed list of the functional and non-functional requirements that the new system must meet.

B Fill the gaps

Use the word bank · 5 marks

external entity | functional | non-functional | measurable | data store

A _____ requirement states what the system must do, while a _____ requirement states how well it must do it. Every requirement should be numbered and _____. In a DFD, a person or organisation outside the system is an _____, and data held by the system is shown as a _____.

C True or false?

Tick one box · 6 marks

	True	False
'Search results must appear in under 3 seconds' is a non-functional requirement.	☐	☐
'The system should be fast' is a good requirement because everyone understands it.	☐	☐
Open questions are useful for exploring how the current system works.	☐	☐
In a DFD, data can flow directly from one data store to another.	☐	☐
Every problem with the current system should lead to at least one requirement.	☐	☐
A context diagram shows every data store in the system.	☐	☐

D Exam-style questions

19 marks

1 Distinguish between a functional and a non-functional requirement, giving an example of each for a boat hire booking system. [4]

2 Explain why the requirement 'The system should be user-friendly' is weak and rewrite it. [3]

3 Give one advantage and one disadvantage of interviewing the client as a fact-finding method. [2]

4 State the purpose of each DFD symbol: external entity, process, data store, data flow. [4]

5 Describe the difference between a context diagram and a level 1 DFD. [2]

6 Explain how a well-written requirements specification helps in the testing and evaluation stages of the project. [4]

E Quick quiz

Circle one answer · 5 marks

1 Which is a functional requirement? [1]

A Results appear in 3 seconds

B The system must calculate the total cost of a booking

C Only staff can log in

D The menu uses a large font

2 Which fact-finding method allows follow-up questions? [1]

A Questionnaire

B Document analysis

C Interview

D Observation

3 What makes a requirement measurable? [1]

A It uses the word 'should'

B It has a clear target that can be tested

C It is written by the client

D It mentions the software

4 In a DFD, what does an open-ended box labelled D1 show? [1]

A Data store

B External entity

C Process

D Data flow

5 Which diagram shows the whole system as a single process? [1]

A Level 1 DFD

B ER diagram

C Gantt chart

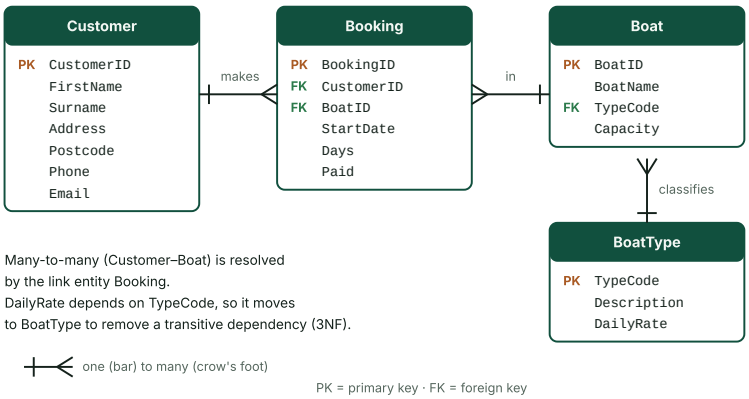
D Context diagram

Design: ER diagrams, normalisation, data dictionary and interfaces

AD03 WORKSHEET

Name	Class	Date	Score

I can: Draw an ER diagram and normalise data to third normal form · Produce a complete data dictionary with validation and keys · Design wireframes, queries and pseudocode algorithms linked to requirements



Use this diagram to help you: An ER diagram for Erne Boat Hire in third normal form: Customer and Boat each have a one-to-many relationship with Booking, and BoatType has a one-to-many relationship with Boat

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Entity relationship diagram**
- ☐ **Foreign key**
- ☐ **Normalisation**
- ☐ **Third normal form**
- ☐ **Data dictionary**
- ☐ **Wireframe**

A A labelled sketch of the layout of a form, report or screen showing its controls and navigation before it is built.

B The process of organising data into related tables to remove redundancy and prevent update, insertion and deletion anomalies.

C A field in one table that is the primary key of another table, used to create a relationship between them.

D A document recording each field's name, data type, size, validation rule, key status and description.

E A table is in 3NF when it is in 2NF and no non-key attribute depends on another non-key attribute.

F A diagram showing the entities in a system and the one-to-one, one-to-many or many-to-many relationships between them.

B Fill the gaps

Use the word bank · 5 marks

third normal form | foreign key | ER diagram | link entity | data dictionary

An _____ shows entities and the relationships between them. A many-to-many relationship is resolved using a _____. Data in _____ has no partial or transitive dependencies. A

_____ links a table to the primary key of another table. The _____ records the type, size and validation of every field.

C True or false?

Tick one box · 6 marks

	True	False
A primary key uniquely identifies each record in a table.	☐	☐
Relational databases implement many-to-many relationships directly without a link table.	☐	☐
A transitive dependency is removed when moving from 2NF to 3NF.	☐	☐
'Range check' is enough detail for a validation rule in a data dictionary.	☐	☐
Wireframes should be annotated with controls, validation and navigation.	☐	☐
Normalisation increases data redundancy.	☐	☐

D Exam-style questions

22 marks

1 Define the term foreign key. [2]

2 Explain why a many-to-many relationship between Customer and Boat must be resolved, and how it is resolved. [3]

3 A Boat table contains BoatID, BoatName, TypeCode and DailyRate, where the daily rate depends on the type of boat. Explain which normal form is not satisfied and how to correct it. [4]

4 State four items that should be recorded in a data dictionary for each field. [4]

5 Describe what should be included in an annotated wireframe for a data entry form. [3]

6 Discuss the importance of the design stage in producing a high-quality database application. [6]

E Quick quiz

Circle one answer · 5 marks

1 Which normal form removes repeating groups? [1]

A First normal form
B Second normal form
C Third normal form
D Unnormalised form

2 How is a many-to-many relationship resolved? [1]

A Delete one entity
B Add a link entity
C Use a larger primary key
D Merge the tables

3 Which is a transitive dependency? [1]

A BookingID → Days
B CustomerID → Surname
C TypeCode → DailyRate in the Boat table
D BoatID → BoatName

4 Which item does NOT belong in a data dictionary? [1]

A Data type
B Field size
C Validation rule
D The client's opinion of the colour scheme

5 What should be designed in pseudocode? [1]

A The ER diagram
B Any code or complex macro
C The Gantt chart
D The title page

Development: building and evidencing the database application

AD04 WORKSHEET

Name	Class	Date	Score

I can: Build tables and relationships that match the design and enforce referential integrity · Create validated forms, parameter and calculated queries, reports and a menu · Produce annotated screenshot evidence that links each feature to the design and requirements

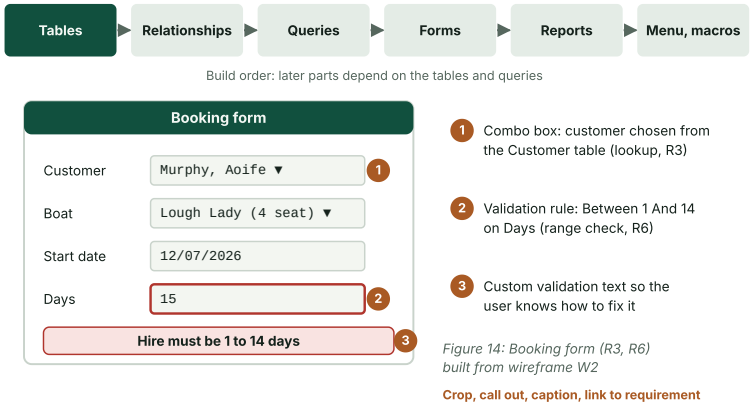


Figure 14: Booking form (R3, R6) built from wireframe W2

Crop, call out, caption, link to requirement

Use this diagram to help you: The order for building a database application, and an annotated screenshot of a booking form showing a lookup, a range check and its error message

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Referential integrity**
- ☐ **Parameter query**
- ☐ **Calculated field**
- ☐ **Input mask**
- ☐ **Macro**
- ☐ **Annotated screenshot**

A A cropped screen capture with labels and a caption explaining what it shows and which requirement and design item it evidences.

B A field in a query, form or report whose value is derived from other fields, such as cost from days and daily rate.

C A stored sequence of actions that automates a task in a database, such as opening a form or printing a report.

D A query that prompts the user to enter a value when it runs and uses that value as its criteria.

E A pattern that controls the format of data entered into a field, such as a postcode or phone number.

F A rule ensuring every foreign key value matches an existing primary key value in the related table, preventing orphan records.

B Fill the gaps

Use the word bank · 5 marks

referential integrity | foreign | parameter query | calculated field | annotated

Tables are linked using primary and _____ keys, and _____ prevents orphan records. A _____ asks the user for a value when it runs. A _____ derives a value such as cost from

other fields. Every feature should be evidenced with an _____ screenshot linked to a requirement.

C True or false?

Tick one box · 5 marks

	True	False
Referential integrity stops a booking being saved for a customer who does not exist.	☐	☐
A calculated field should be stored as a separate field in the table.	☐	☐
A parameter query can be reused for different dates without editing it.	☐	☐
Validation guarantees that the data entered is correct.	☐	☐
Showing a problem you found and fixed is useful development evidence.	☐	☐

D Exam-style questions

19 marks

1 Explain what is meant by referential integrity and why it is important in a booking database. [3]

2 Describe what a parameter query is and give an example from a boat hire system. [3]

3 Explain why the total cost of a booking should be produced by a calculated field rather than stored in the Booking table. [2]

4 Give three validation checks that could be applied to fields in a Booking form, stating the field and the rule. [3]

5 Describe two features of a good annotated screenshot in a development section. [4]

6 A candidate enforces referential integrity with cascade delete between Customer and Booking. Discuss whether this is appropriate. [4]

E Quick quiz

Circle one answer · 5 marks

1 What does enforcing referential integrity prevent? [1]

A Duplicate field names
B Orphan records
C Slow queries
D Large file sizes

2 Which criteria creates a parameter query? [1]

A > 10
B Between 1 And 14
C [Enter surname:]
D Is Null

3 Which is a calculated field? [1]

A Cost: [Days]*[DailyRate]
B Surname
C BookingID
D Postcode

4 What is the best order to build in? [1]

A Reports, forms, tables
B Menu, queries, tables
C Forms, tables, relationships
D Tables, relationships, queries, forms, reports

5 Which caption is the strongest evidence? [1]

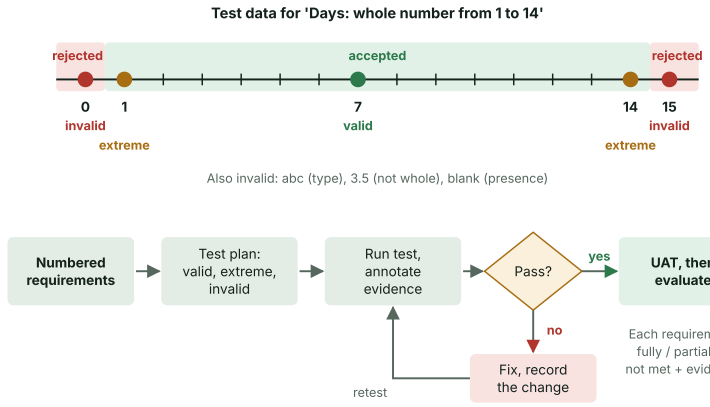
A Screenshot 4
B My form
C Figure 14: Booking form (R3, R6) from wireframe W2
D Done

Testing and evaluation against the requirements

AD05 WORKSHEET

Name	Class	Date	Score

I can: Write a test plan that covers every requirement with valid, invalid and extreme data · Present annotated test evidence including fixes, retests and user acceptance testing · Evaluate each requirement as fully, partially or not met and suggest justified improvements



Use this diagram to help you: Choosing valid, extreme and invalid test data for a 1 to 14 range, and the cycle from requirements through testing, fixing and retesting to user acceptance testing and evaluation

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Test plan**
- ☐ **Valid data**
- ☐ **Extreme data**
- ☐ **Invalid data**
- ☐ **User acceptance testing**
- ☐ **Evaluation**

A Data that is outside the acceptable range or of the wrong type, which the system should reject.

B Data at the boundaries of the acceptable range, which the system should accept; also called boundary data.

C Normal data within the acceptable range that the system should accept.

D A judgement of how well the finished system meets each requirement, supported by evidence, with justified improvements.

E Testing by the client or end users with realistic tasks to confirm the system meets their requirements.

F A planned list of tests, each with a number, requirement, purpose, test data, data type and expected result.

B Fill the gaps

Use the word bank · 5 marks

test plan | user acceptance | extreme | invalid | met

A _____ lists each test with its data and expected result. Data at the edge of the acceptable range is _____ or boundary data, and data that should be rejected is _____ data. Testing by the client is called _____ testing. In the evaluation each requirement is judged fully, partially or not _____.

C True or false?

Tick one box · 6 marks

	True	False
For a range of 1 to 14, the value 14 is extreme data and should be accepted.	☐	☐
Only validation needs to be tested in a database project.	☐	☐
A failed test that is fixed and retested is useful evidence.	☐	☐
'The system is good' is an evaluative statement.	☐	☐
User acceptance testing is carried out by the client or end users.	☐	☐
Expected results are written after the tests are carried out.	☐	☐

D Exam-style questions

22 marks

1 A field must hold a whole number of days from 1 to 14. Give one example each of valid, extreme and invalid test data. [3]

2 State four columns that should appear in a test plan. [4]

3 Explain why a test plan should be mapped to the numbered requirements. [2]

4 Describe what is meant by user acceptance testing and why it is carried out. [3]

5 A candidate writes: 'The search works well and the client liked it.' Explain why this is a weak evaluation and rewrite it. [4]

6 Evaluate the importance of testing with extreme and invalid data, as well as valid data, in a database application. [6]

E Quick quiz

Circle one answer · 5 marks

1 For 'Days from 1 to 14', which value is extreme data? [1]

A 7 B 14
C 15 D abc

2 Which is invalid data for 'Days from 1 to 14'? [1]

A 0 B 1
C 7 D 14

3 Who carries out user acceptance testing? [1]

A The moderator B The developer only
C The client or end users D CCEA

4 What should support each evaluation judgement? [1]

A Personal opinion B The length of the section
C The software used D Evidence such as test numbers and feedback

5 A requirement that works for most cases but not all is... [1]

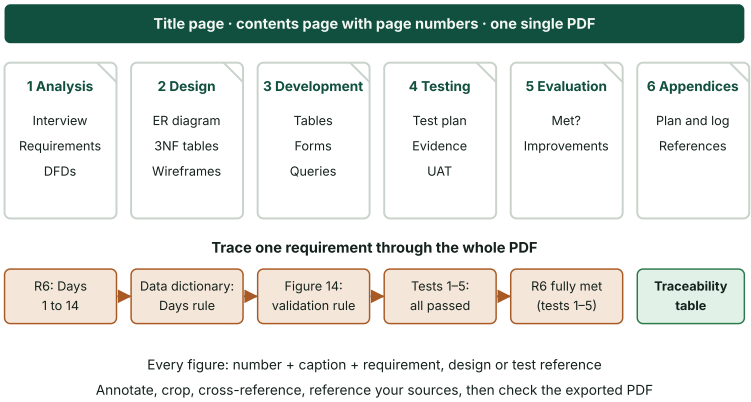
A Partially met B Fully met
C Not met D Invalid

Producing the portfolio

AD06 WORKSHEET

Name	Class	Date	Score

I can: Structure a PDF portfolio with a contents page, numbered sections and figures · Annotate and cross-reference screenshots so evidence is easy to find · Reference sources correctly and check the portfolio against a final checklist



Use this diagram to help you: The structure of the single PDF portfolio in six sections, and how one requirement is traced through every section by cross-references

A Key terms

Write the letter of the matching definition · 4 marks

- ☐ **Portfolio**
- ☐ **Cross-reference**
- ☐ **Traceability table**
- ☐ **Referencing**

A Acknowledging the sources used in a piece of work in a consistent format, so it is clear which work is the candidate's own.

B A reference from one part of the portfolio to another, such as a test number or figure number, so evidence can be traced.

C A table linking each requirement to its design, development, test and evaluation evidence.

D The single PDF document containing all the candidate's evidence of analysis, design, development, testing and evaluation.

B Fill the gaps

Use the word bank · 5 marks

caption | contents | cross-reference | PDF | references

The portfolio is submitted as a single _____. A _____ page with page numbers helps the moderator navigate. Every screenshot needs a figure number and a _____. Requirement and test numbers are used to _____ evidence. All sources are listed in the _____ section.

C True or false?

Tick one box · 5 marks

	True	False
The portfolio should include a contents page with page numbers.	☐	☐

Copying textbook theory into the portfolio adds to your evidence.	☐	☐
Screenshots should be cropped so that their text is readable.	☐	☐
You only need to reference sources you quoted word for word.	☐	☐
You should check the PDF after exporting it.	☐	☐

D Exam-style questions

14 marks

1 State three features that make a PDF portfolio easy for a moderator to navigate. [3]

2 Describe how a screenshot should be presented as evidence of development. [3]

3 What is a traceability table and why is it useful? [2]

4 Explain why all sources, including tutorials and templates, must be referenced in the portfolio. [2]

5 Give two common presentation mistakes in coursework portfolios and how to avoid them. [4]

E Quick quiz

Circle one answer · 5 marks

1 How is the A2 2 portfolio submitted? [1]

A As a single PDF
B As the database file only
C As a printed folder only
D As a video

2 What should every screenshot have? [1]

A A border
B A figure number and caption
C A background colour
D A title page

3 What does a traceability table show? [1]

A The project budget
B The order of slides

C Each requirement's design, development, test and evaluation references

D A list of software

4 Which must be acknowledged? [1]

A Only books

B Only quotations

C Nothing

D Tutorials, websites, templates and permitted AI use

5 When should the contents page be checked? [1]

A Before starting analysis

B After the final export

C Only if the moderator asks

D Never